

MAT 344 HW07

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Problem 1 (p.171#5). Show that the mean value formula remains valid for $u = \log |1 + z|$, $z_0 = 0$, $r = 1$, and use this fact to compute

$$\int_0^\pi \log \sin \theta d\theta.$$

Solution. Observe that $\log |1 + z|$ is harmonic in the open disk $B_1(0)$ because it is the real part of the holomorphic function $\log(1 + z)$ defined in the simply connected domain $B_1(0)$, so certainly

$$0 = u(0) = \frac{1}{2\pi} \int_0^{2\pi} \log |1 + re^{i\theta}| d\theta$$

for $0 \leq r < 1$. We need to show that

$$\lim_{r \rightarrow 1^-} \frac{1}{2\pi} \int_0^{2\pi} \log |1 + re^{i\theta}| d\theta = \frac{1}{2\pi} \int_0^{2\pi} \log |1 + e^{i\theta}| d\theta.$$

Consider the function $g: [0, 2\pi] \rightarrow \mathbb{R}$ given by

$$g(\theta) = \max \left(1, \frac{2}{\sqrt{|\pi - \theta|}} \right).$$

Clearly g is integrable on $[0, 2\pi]$ because $\frac{1}{\sqrt{x}}$ is integrable on any bounded subset of $\mathbb{R}$. Moreover, we can bound

$$\frac{|\pi - \theta|}{2} < |\sin \theta| \leq |1 + e^{i\theta}|$$

for θ sufficiently close to π (since $|\sin \theta|$ is the distance from $1 + e^{i\theta}$ to the real axis), so

$$|\log(1 + e^{i\theta})| \leq \frac{1}{\sqrt{|1 + e^{i\theta}|}} \leq \frac{2}{\sqrt{|\pi - \theta|}}.$$

We can note that for a given θ , $|1 + re^{i\theta}|$ is maximized over $r \in [0, 1]$ at either $r = 0$ or $r = 1$ because the squared distance is quadratic in r , hence is a convex function of r , so $g(\theta)$ is an upper bound for $|\log |1 + re^{i\theta}||$. for all $r \in [0, 1]$ and $\theta \in [0, 2\pi]$, hence the dominated convergence theorem implies the desired convergence.

Using the identity $1 + \cos \theta = 2 \cos^2(\frac{\theta}{2})$ and the integral computed above,

$$\begin{aligned} 0 &= \int_0^{2\pi} \log |1 + e^{i\theta}| d\theta = \int_0^{2\pi} \log(\sqrt{2 + 2 \cos \theta}) d\theta \\ &= \int_0^{2\pi} \log \left(2 \left| \cos \left(\frac{\theta}{2} \right) \right| \right) d\theta = 2 \int_0^\pi \log \left(2 \cos \frac{\theta}{2} \right) d\theta \\ &= \pi \log 2 + \int_0^\pi \log \left(\cos \frac{\theta}{2} \right) d\theta = \pi \log 2 + \int_0^\pi \log \left(\sin \frac{\theta}{2} \right) d\theta \\ &= \pi \log 2 + 2 \int_0^{\pi/2} \log(\sin u) du = \pi \log 2 + \int_0^\pi \log(\sin u) du, \end{aligned}$$

so the integral in the last line, which is what we want to find, is equal to $\boxed{-\pi \log 2}$. □

Problem 2 (p.171#6). If $f(z)$ is analytic in the whole plane and if $z^{-1} \operatorname{Re} f(z) \rightarrow 0$ when $z \rightarrow \infty$, show that f is a constant.

Solution. Let u be the real part of $g(z) = f(z) - f(0)$, so that equation (66) says

$$g(z) = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{\zeta + z}{\zeta - z} \cdot u(\zeta) \frac{d\zeta}{\zeta} + iC$$

for some constant C , if $|z| < R$. We have that $\lim_{z \rightarrow \infty} \frac{u(z)}{z} = 0$ and $\lim_{z \rightarrow 0} \frac{g(z)}{z} = f'(0)$, implying that $\frac{u(z)}{z}$ is bounded everywhere, so suppose that $|u(z)| \leq M|z|$ for some constant $M > 0$. Taking $R = 2|z|$, we find

$$|f(z)| \leq \frac{1}{2\pi} C + \int_{|\zeta|=R} \left| \frac{Re^{i\theta} + z}{Re^{i\theta} - z} \right| u(Re^{i\theta}) d\theta \leq C + 3MR = C + 6M|z|,$$

noting that $|Re^{i\theta} + z| \leq 3|z|$ and $|Re^{i\theta} - z| \geq |z|$. We showed in a previous homework that this implies f is a linear polynomial $f(z) = az + b$. Then, if $a \neq 0$,

$$a = \lim_{r \rightarrow \infty} \frac{r|a|^2 + b}{r\bar{a}} = \lim_{r \in \mathbb{R}} \frac{u(r\bar{a}) + \operatorname{Im} b}{r\bar{a}} = \lim_{z \rightarrow \infty} \frac{u(z)}{z},$$

so the right-hand limit is nonzero, which contradicts the given information; thus a is zero and $f(z) \equiv b$. $\square$

Problem 3 (p.174#3). If $f(z)$ is analytic in $|z| \leq 1$ and satisfies $|f| = 1$ on $|z| = 1$, show that $f(z)$ is rational.

Solution. Let L be a fractional linear transformation biholomorphically mapping the closed disk $\overline{\Delta}$ to the closed upper half plane $\overline{\mathbb{H}}$ (where $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ and the closure is taken in the Riemann sphere, so that $\partial\mathbb{H} = \mathbb{R} \cup \infty$), such as $L(z) = i\frac{1+z}{1-z}$. Then, consider $g = LfL^{-1} : \overline{\mathbb{H}} \rightarrow \overline{\mathbb{H}}$. Assuming f is nonconstant, the maximum principle tells us that $f(\Delta) \subseteq \Delta$ and we are given that $f(\partial\Delta) \subseteq \partial\Delta$. This translates to $g(\mathbb{H}) \subseteq \mathbb{H}$ and $g(\partial\mathbb{H}) \subseteq \partial\mathbb{H}$, meaning in particular that the poles and zeroes of g are all real; thus

$$g(z) = \frac{(z - z_1) \cdots (z - z_n)}{(z - w_1) \cdots (z - w_m)} h(z)$$

for $z_i, w_i \in \mathbb{R}$ and $h \in \mathcal{O}(\overline{\mathbb{H}})$ a nowhere zero and nowhere singular holomorphic function with $h(\partial\mathbb{H}) \subseteq \mathbb{R} - \{0\}$.

I claim that h must be a constant function. Let $k = L^{-1}hL : \overline{\Delta} \rightarrow \overline{\Delta}$; since $h(\partial\mathbb{H}) \subseteq \mathbb{R} - \{0\}$, $k(\partial\Delta) \subseteq \partial\Delta - \{L^{-1}(0) = -1\}$. It is easy to see (e.g. by stereographic projection) that $\partial\Delta - \{-1\}$ is homeomorphic to the contractible space $\mathbb{R}$, so $k|_{\partial\Delta} : \partial\Delta \rightarrow \partial\Delta$ is nullhomotopic, meaning the winding number of k around any interior point of $\partial\Delta$ is zero. By the argument principle, we therefore find that $k(\Delta) \subseteq \partial\Delta$, so the fact that nonconstant holomorphic functions are open maps tells us that k , hence h , is constant; in fact $h(z) \equiv c$ for some nonzero real c .

We conclude that

$$LfL^{-1}(z) = g(z) = c \cdot \frac{(z - z_1) \cdots (z - z_n)}{(z - w_1) \cdots (z - w_m)}$$

is rational, so certainly the composition of rational functions $g = L^{-1}fL$ is also rational. $\square$

Problem 4 (p.174#4). Use (66) to derive a formula for $f'(z)$ in terms of $u(z)$.

Solution. Equation (66) says that for $z, w \in \mathbb{C}$ with $|z|, |w| < R$,

$$\begin{aligned} \frac{f(z) - f(w)}{z - w} &= \frac{1}{z - w} \cdot \frac{1}{2\pi i} \int_{|\zeta|=R} \left(\frac{\zeta + z}{\zeta - z} - \frac{\zeta + w}{\zeta - w} \right) \cdot u(\zeta) \frac{d\zeta}{\zeta} \\ &= \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{1}{z - w} \cdot \frac{2(z - w)\zeta}{(\zeta - z)(\zeta - w)} \cdot u(\zeta) \frac{d\zeta}{\zeta} = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{2u(\zeta)}{(\zeta - z)(\zeta - w)} d\zeta. \end{aligned}$$

Take $\delta > 0$ such that $|w| + \delta < R$ and z such that $|w - z| < \delta$. The magnitude of the expression

$$\frac{1}{2\pi i} \int_{|\zeta|=R} \frac{2u(\zeta)}{(\zeta - z)(\zeta - w)} - \frac{2u(\zeta)}{(\zeta - w)^2} d\zeta = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{2u(\zeta)}{\zeta - w} \cdot \frac{z - w}{(\zeta - z)(\zeta - w)} d\zeta$$

can be bounded above by

$$\frac{1}{2\pi} \int_0^{2\pi} \left| \frac{2u(\zeta)}{\zeta - w} \right| \cdot \left| \frac{z - w}{(\zeta - z)(\zeta - w)} \right| \cdot R d\theta$$

where of course $\zeta = Re^{i\theta}$. Since $|\zeta - a| \geq R - |a|$ for any $a \in \mathbb{C}$ (noting that $|\zeta| = R$) and $|z|$ and $|w|$ are both bounded above by $|w| + \delta$, we have

$$\left| \frac{z - w}{(\zeta - z)(\zeta - w)} \right| < \frac{\delta}{(R - |z|)(R - |w|)} < \frac{\delta}{(R - |w| - \delta)^2},$$

so

$$\frac{1}{2\pi} \int_0^{2\pi} \left| \frac{2u(\zeta)}{\zeta - w} \right| \cdot \left| \frac{z - w}{(\zeta - z)(\zeta - w)} \right| \cdot R d\theta < \frac{\delta}{(R - |w| - \delta)^2} \cdot \frac{R}{2\pi} \int_0^{2\pi} \left| \frac{u(\zeta)}{\zeta - w} \right| d\theta.$$

The final expression goes to zero with δ , meaning

$$\lim_{z \rightarrow w} \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{2u(\zeta)}{(\zeta - z)(\zeta - w)} - \frac{2u(\zeta)}{(\zeta - w)^2} d\zeta = 0 \iff \boxed{f'(w) = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{2u(\zeta)}{(\zeta - w)^2} d\zeta}. \quad \square$$

Problem 5 (p.174#5). If $u(z)$ is harmonic and $0 \leq u(z) \leq Ky$ for $y > 0$, prove that $u = ky$ with $0 \leq k \leq K$.

Solution. Assume that u is defined on the closed half plane $\overline{\mathbb{H}}$. The given condition implies $u(z) = 0$ on the real line, so u admits an extension $\tilde{u}: \mathbb{C} \rightarrow \mathbb{R}$ with $\tilde{u}(\bar{z}) = -\tilde{u}(z)$ on all of $\mathbb{C}$. Let v be a conjugate harmonic function for $\tilde{u}$, so $f = \tilde{u} + iv$ is entire. By the previous problem, for a given z and large enough R ,

$$f'(z) = \frac{1}{2\pi i} \int_{|\zeta|=R} \frac{2u(\zeta)}{(\zeta - z)^2} d\zeta.$$

By the given condition, $|u(z)| \leq K \cdot \text{Im } z \leq K|z|$ for all z , so

$$|f'(z)| \leq \frac{1}{2\pi} \int_0^{2\pi} \left| \frac{2u(Re^{i\theta})}{(Re^{i\theta} - z)^2} \right| \cdot R d\theta \leq \frac{1}{2\pi} \int_0^{2\pi} \frac{R^2 K}{|Re^{i\theta} - z|^2} d\theta = \frac{1}{2\pi} \int_0^{2\pi} \frac{K}{|e^{i\theta} - \frac{z}{R}|^2} d\theta.$$

For each ε , there exists R_ε such that $|e^{i\theta} - \frac{z}{R}|$ is within ε of 1 when $R > R_\varepsilon$, so $|f'(z)|$ is bounded above by $\frac{K}{1+\varepsilon}$ for sufficiently large R . By Liouville's theorem f' is therefore constant, and more specifically we have bounded $|f'(z)| \leq K$.

Finally, if $f(z) = az + b$ for constant a and b , then the real part of $f(z)$ is zero whenever z is real. This forces a and b to be purely imaginary: by taking $z = 0$, b must have zero real part, and then taking $z = 1$ tells us that a has zero real part. Thus $f(z) = (kz + \ell)i$ for $0 \leq k \leq K$ and real ℓ , so $\tilde{u}(z) = ky$ as desired. $\square$